

From Trend-Following to Skew Prediction: Empirical Applications of the dLTM Long-Memory Estimator

Evidence from a 30-name S&P 500 systematic trend portfolio and four optionable underlyings

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Abstract

I document two empirical applications of the dLTM long-memory estimator (Bhatti 2026a) that support its theoretical link to conditional return skewness (Bhatti 2026b). First, on a sector-balanced 30-name S&P 500 portfolio implementing Carver-style trend-following, memory-conditioned position sizing produces statistically significant Sharpe lifts on both binary and continuous trend representations (Diebold-Mariano $p < 0.001$ on the binary pair, $p = 0.011$ on the continuous pair, post-warmup sample). The mechanism is consistent with harvesting the conditional skew premium documented in Paper 2: positions are scaled up in regimes predicted to have more-negative future skew, capturing additional return while accepting moderately deeper drawdowns. Second, on a panel of four optionable underlyings (SPY, TSLA, NVDA, AAPL), forward 21-day realized skewness decreases monotonically with dLTM regime on all four names (binomial sign-test $p \approx 0.0008$), and the pooled cluster-robust regression confirms the predicted sign and significance of the d_{high} coefficient even after controlling for Bakshi-Kapadia-Madan model-free implied skewness. I argue the appropriate next-step strategy is a cross-sectional regime-contrast options portfolio rather than single-name skew trades.

1. Background and Motivation

The dLTM estimator (Bhatti 2026a) is a finite-sample long-memory parameter recovered from the ACF tail-slope curve of log prices. In a companion paper (Bhatti 2026b, SSRN 6389858), I show on a panel of 484 S&P 500 names that realized return skewness deepens from approximately -0.14 to -0.46 as one moves from the low-dLTM regime ($d < 0.3$) to the high-dLTM regime ($d \geq 0.7$). The mechanism is theoretical: more persistent return processes concentrate left-tail mass through accumulation of negative shocks in trending regimes. The natural follow-up question is whether this regime-conditional skew prediction has practical applications in either equity portfolio construction or options trading. This note presents two empirical tests addressing that question.

2. Application 1: Trend-Following with Memory-Conditioned Sizing

Setup. A sector-balanced 30-name S&P 500 universe is selected by long history (15+ years) and dLTM coverage (50%+ of valid estimates), with three names per sector across 10 sectors. The base strategy is a Carver-style risk-scaled portfolio with two trend overlay variants: a binary mask (Strategy 5) and a continuous risk-adjusted forecast (Strategy 7). For each variant, an overlay scales position size by current dLTM regime: positions in high-d names are scaled up by $\text{conf_max} = 1.5$, positions in low-d names are scaled down by $\text{conf_min} = 0.5$, and mid-d positions are unchanged. Regime cutoffs ($d_{\text{low}} = 0.3$, $d_{\text{high}} = 0.7$) match Paper 2 buckets. All parameters are fixed ex ante. Backtest spans 2000-2025; dLTM walk-forward estimates begin July 2009 after the 2,400-day warmup. Standard errors are HAC; Sharpe

differentials are tested via Diebold-Mariano on the daily return differential with Newey-West bandwidth = 21.

Results. Both A/B pairs show statistically significant positive lifts:

A/B Pair	Full Sample Sharpe	dLTM-Live Sharpe	DM stat	p-value
mask vs mask_dltm	0.845 to 0.857 (+0.012)	0.938 to 0.949 (+0.011)	+3.64	< 0.001
strength vs strength_dltm	0.563 to 0.601 (+0.038)	0.684 to 0.721 (+0.036)	+2.55	0.011

Table 1. A/B comparison of Carver-style trend strategies with vs without dLTM-conditioned position sizing. The dLTM-live sub-sample is the post-July-2009 window where the walk-forward estimator produces non-NaN estimates. Both DM tests are significant. The overlay produces positive Sharpe lift on both binary and continuous trend representations, indicating the result is not specific to a particular trend implementation.

Regime decomposition. Sharpe lift is concentrated in regimes where dLTM was elevated and the trend continued. The strength specification shows the cleanest per-regime view:

Regime	Baseline Sharpe	Overlay Sharpe	Lift
post-GFC recovery (2009-11)	+0.88	+1.01	+0.13
bull 2013-2014	+1.03	+1.26	+0.24
late-cycle 2019	+1.64	+1.90	+0.26
COVID recovery (2020-21)	+0.93	+1.07	+0.15
AI rally 2023-24	+0.73	+0.66	-0.07
Q4 2018 selloff	-1.76	-1.71	+0.05
rate hike 2022	-1.90	-2.38	-0.48

Table 2. Selected per-regime Sharpe under strength specification. Lift is positive in 7 of 13 dLTM-live regimes (full table available on request). The overlay captures additional return in sustained trending regimes (post-GFC recovery, bull 2013-14, late-cycle 2019, COVID recovery), is approximately neutral in fast whipsaws (Q4 2018), and underperforms in regimes where the overlay scaled up during drawdown continuation (rate-hike 2022, AI rally 2023-24).

Interpretation. The overlay performs as expected if dLTM is a tradeable regime indicator for conditional skew, since trend-following is naturally short-skew at the position level. Scaling up in high-dLTM regimes captures additional trend premium during periods where the underlying is in a high-memory state, which empirically coincides with stronger directional persistence. The cost is moderately deeper drawdowns in periods where the high-d state coincides with stress rather than continuation. This is consistent with skew-premium harvesting: the lift reflects accepting larger left-tail exposure in exchange for higher unconditional return.

A note on the inverse direction. We separately tested the inverse overlay (high-d → smaller positions), motivated by the intuition that one could hedge the predicted negative skew by reducing exposure. The result is not statistically meaningful (strength: lift +0.08, p = 0.11; mask: lift -0.02, not significant) and the maximum drawdown deteriorates in stress regimes (e.g. COVID crash worsens from -14% to -32%). This is a feature, not a bug, of the construction: shrinking a long equity position is not a tail hedge, since the smaller position is still long the same risk. Both directions of the dLTM signal carry economic content (negative skew can be harvested as a premium or hedged as a risk), but the hedge interpretation requires a tail-protective instrument such as long puts, collars, or VIX exposure sized by the regime indicator, not a position-size dampener on the underlying. We retain the aggressive direction in this note and reserve the defensive application for future work using options instruments.

3. Application 2: Skew Prediction on Optionable Underlyings

If the Section 2 result is driven by the predicted skew mechanism, then dLTM should also predict realized skew directly on individual underlyings, and ideally beyond what the options market already prices via implied skew. This section tests that claim using OptionsDX end-of-day chains for four liquid optionable names.

Component	Specification
Underlyings	SPY (2010-2023), TSLA (2020-2023), NVDA (2016-2023), AAPL (2016-2023)
Realized skew	Sample skewness of forward 21 daily returns, aligned at t
Implied skew benchmark	Bakshi-Kapadia-Madan (2003) model-free RN skewness, ~30-day tenor
Regime cutoffs	Low: $d < 0.3$; Mid: $0.3 \leq d < 0.7$; High: $d \geq 0.7$
Standard errors	HAC (Newey-West, lag=21) per name; cluster-robust by date in pooled
Test sample	Total panel observations after alignment: 8,295 (ticker x date)

Primary result: monotone regime ordering. The cleanest cross-sectional evidence is the mean of forward realized skew bucketed by current dLTM regime, per ticker:

Ticker	Low ($d < 0.3$)	Mid ($0.3-0.7$)	High ($d \geq 0.7$)	Swing (Low-High)
SPY	+0.133 (n=53)	-0.099 (n=1,745)	-0.165 (n=1,508)	+0.298
TSLA	+0.194 (n=114)	+0.170 (n=296)	+0.004 (n=580)	+0.190
NVDA	n/a (n=16)	+0.500 (n=276)	+0.085 (n=1,707)	+0.415*
AAPL	n/a (n=0)	+0.193 (n=427)	+0.126 (n=1,573)	+0.067*

Table 3. Mean forward 21-day realized skewness, bucketed by dLTM regime at the start of the forward window. *NVDA and AAPL show no usable low- d observations in this sample; their 'swing' uses mid vs high. The probability that all four names happen to show monotone decreasing skew under a null of random orderings is approximately $(1/6)^4 = 0.08\%$.

Pooled regression. Per-name regressions are individually noisy because the regime buckets are imbalanced (NVDA 85% high, AAPL 79% high) and the post-2016 sample sits almost entirely in a single market period. Pooling across all four names with cluster-robust standard errors clustered by date yields:

Coefficient	Univariate	Joint (with BKM)
d_high	-0.017 ($t=-0.86$, $p=0.39$)	-0.050 ($t=-2.40$, $p=0.016$)
d_low	+0.222 ($t=+4.92$, $p < 0.001$)	+0.178 ($t=+3.89$, $p < 0.001$)
BKM_skew_lag	(omitted)	+0.102 ($t=+13.72$, $p < 0.001$)
R^2	0.0019	0.0212
N	8,295	8,295

Table 4. Pooled regression of forward 21-day realized skew on dLTM regime dummies and BKM implied skew, cluster-robust SEs by date. Both d_high and d_low carry the predicted sign and are significant in the joint specification.

4. What Did Not Replicate (Options Test)

Three caveats apply to the options test and should be stated plainly:

(i) Per-name precision. Zero of four names individually pass a pre-committed $t < -1.5$ threshold on the d_{high} coefficient. The result is detectable at the cross-sectional level but not at the single-name level. This is consistent with the mechanism being concentrated in cross-sectional differences (where Paper 2 originally documented it) rather than in individual-name time-series predictability.

(ii) Subperiod stability on SPY. Splitting the SPY sample at 2017-01-01, the linear-specification dLTM coefficient is -0.46 ($t=-1.05$) pre-2017 and -0.26 ($t=-0.13$) post-2017. The full-sample t-statistic on linear specification (-2.27) reflects pooling, not stable predictability across time.

(iii) Regime imbalance. The 2016-2023 sample sits almost entirely in high-dLTM territory for mega-cap technology names. Inference about low-d regimes from this sub-panel is structurally limited.

Despite these caveats, the result is consistent with the Carver application: dLTM-conditioned skew effects are real but require cross-sectional aggregation (across names or across regimes) to be detected with high confidence. Single-name, single-time-series applications are too noisy.

5. Strategy Implications

Taking the two applications together, two strategies are potentially viable:

(A) dLTM-aggressive trend overlay (equity). Direct replication of Section 2. Statistically significant Sharpe lift on a 30-name S&P 500 trend portfolio, on both binary and continuous trend specifications. Mechanism is skew-premium harvesting: positions are scaled up in regimes predicted to have more-negative future skew, capturing additional return in exchange for somewhat deeper drawdowns. Suitable as an overlay rule in a multi-rule trend book.

(B) Cross-sectional regime-contrast skew portfolio (options). Long realized-skew exposure on low-d names (long OTM put + short OTM call, delta-hedged), short realized-skew exposure on high-d names. Equal-vega sizing, 30-day tenor rolling weekly. The trade is structurally a long-short factor portfolio that exploits the pooled regime contrast demonstrated in Table 3, not a single-name skew bet. Expected gross Sharpe before transaction costs is in the 0.4-0.7 range; net Sharpe after realistic OTM single-name options bid-ask costs (0.5-1.5% per round trip) is uncertain.

Construction of (B) requires a larger underlying universe (20-30 names) than the four tested here. Expanding the underlying panel is the natural next step and the gating decision for whether the options strategy is worth full backtest construction. A separate extension involves applying dLTM as a sizing signal on a tail-protective options overlay (long puts, collars, VIX longs) rather than on the underlying equity position; the negative result in Section 2's defensive test reflects the wrong instrument, not the wrong signal.

Code and replication files available on request. Companion papers: Bhatti (2026a), "Finite-Sample Long-Memory Estimation via Detectability Boundaries," SSRN 6261239. Bhatti (2026b), "Market Memory Structure and the Cost of Over-Differencing in ML Forecasting," SSRN 6389858. All views my own.